Erie 2024 Nearshore Rotifers

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2026-10-20

Summer NS-OS Rotifer Genera Average Biomass Barplot

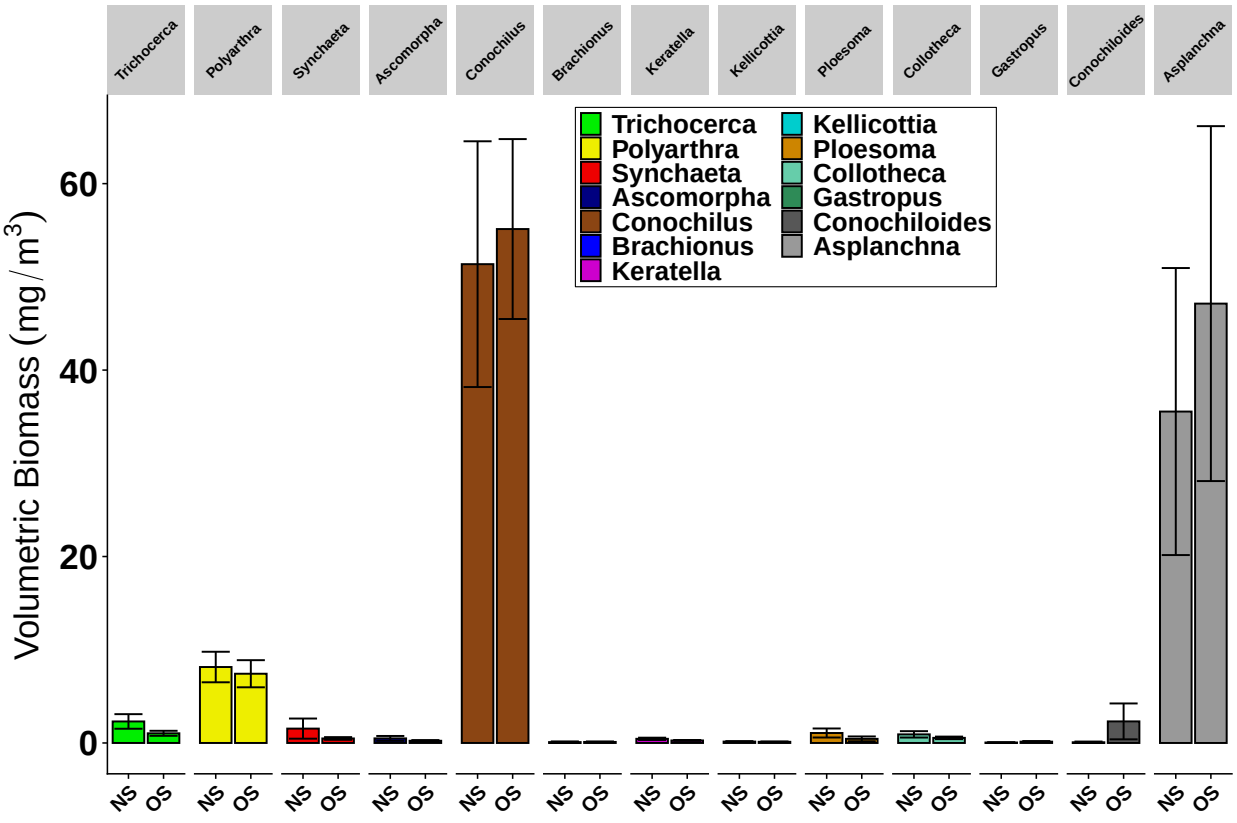


Figure 1: Figure 1A. Summer Rotifer genera nearshore (NS) vs. offshore (OS) average biomass (mg dry wt per m^3). Error bars represent standard errors of the means. Taxa groups are identified in legend by color.

Figure 1A shows the volumetric biomass of the 13 most commonly encountered taxa between the nearshore and offshore samples for Lake Erie in August 2024. With several exceptions, most rotifer taxa exhibited higher average biomass values in the offshore samples, suggesting that these waters supported higher densities and perhaps larger taxa compared to shallower habitats. Several exceptions to this were *Trichocerca*, which had a higher biomass in the nearshore vs. offshore, and can be attributed to the higher nutrient concentrations found in nearshore waters supporting a greater density of these eutrophic-loving taxa (Gannon and Stemberger 1978; Sládecek, 1983). *Synchaeta* and *Ploesoma* showed a slight preference for the nearshore waters while *Conochiloides* showed a preference for the offshore waters. The remaining taxa *Polyarthra*, *Ascomorpha*, *Brachionus*, *Keratella*, etc, exhibited very similar biomass values between near- and offshore waters. These

insights suggest that despite potentially higher food concentrations in nearshore habitats, only several taxa gain a clear advantage in these conditions.

Offshore Only: Finlinia Nearshore Only: Lepadella, Monostyla, Pompholyx, Trichotria

18 rotifer genera in the summer nearshore; 39 speccodes (species) Ascomorpha Asplanchna Brachionus Collotheca Conochiloides Conochilus Gastropus Kellicottia Keratella Lepadella Monostyla Ploesoma Polyarthra Pompholyx Rotifer Bdelloid Synchaeta Trichocerca Trichotria

15 rotifer genera in the summer offshore; 32 speccodes (species) Ascomorpha Asplanchna Brachionus Collotheca Conochilus Filinia Gastropus Kellicottia Keratella Ploesoma Polyarthra Rotifer Bdelloid Synchaeta Trichocerca

References:

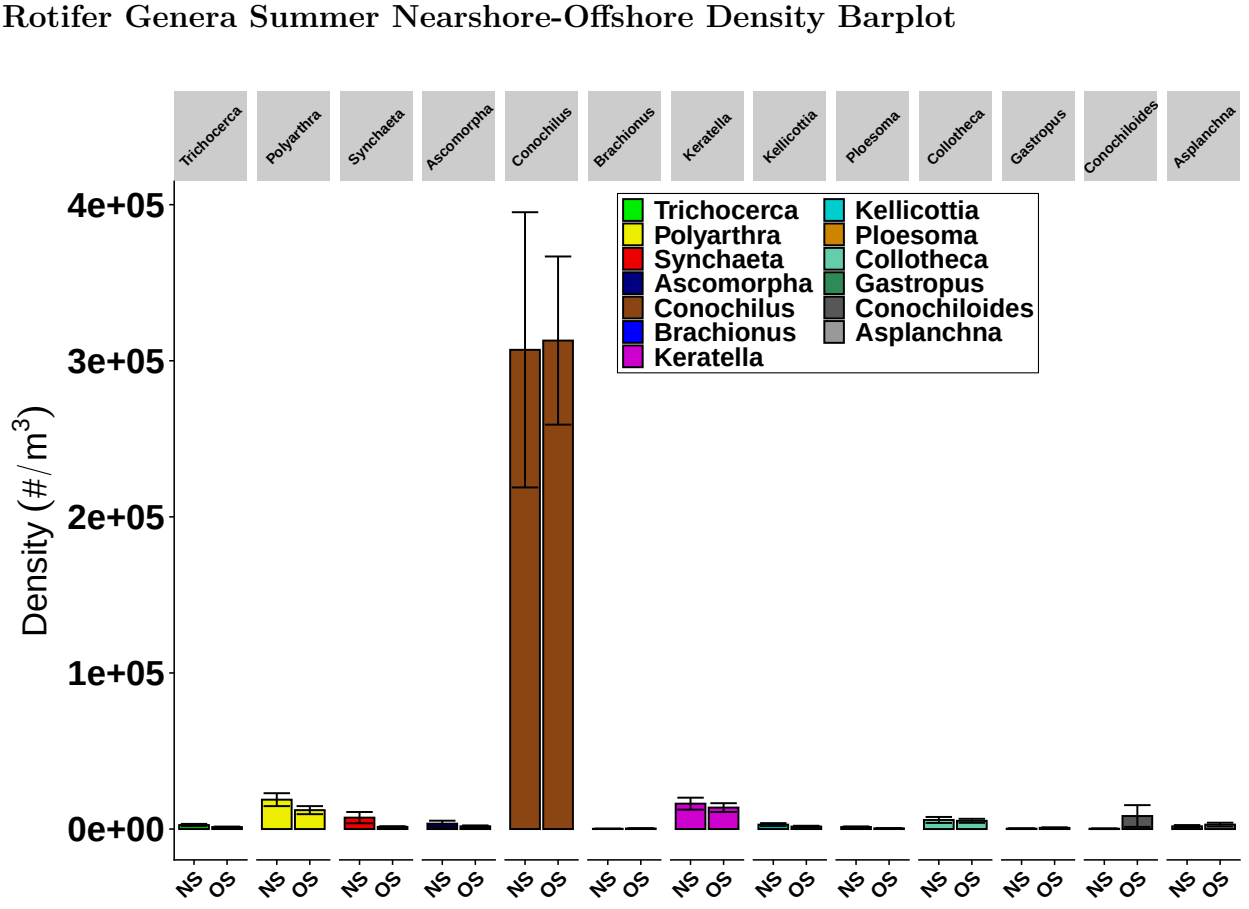


Figure 2: Figure 1B. Summer Rotifer genera nearshore (NS) vs. offshore (OS) average density (# per m^3). Error bars represent standard errors of the means. Taxa groups are identified in legend by color.

Figure 1B shows the density of the 13 most commonly encountered taxa between the nearshore and offshore samples for Lake Erie in August 2024. With several exceptions, most rotifer taxa exhibited higher average density values in the nearshore samples, which is in contrast to average biomass values, indicating that smaller specimens or species are found in shallower habitats. Several taxa (*Trichocerca*, *Polyarthra*, *Synchaeta*, *Ascomorpha*) showed higher densities in the nearshore vs. offshore waters. *Conochiloides* was the only taxa that showed a clear preference for the offshore waters. The remaining taxa *Brachionus*, *Keratella*, *Kellicottia*, etc, exhibited very similar density values between near- and offshore waters. These insights suggest that despite potentially higher food concentrations in nearshore habitats, only several taxa gain an advantage

in these conditions. Additionally, the contrasting findings that densities for several taxa are higher in the nearshore while biomass is similar or higher in the offshore, indicates there is a different rotifer community between the sample types.

Rotifer Genera Spring Nearshore-Offshore Biomass Barplot

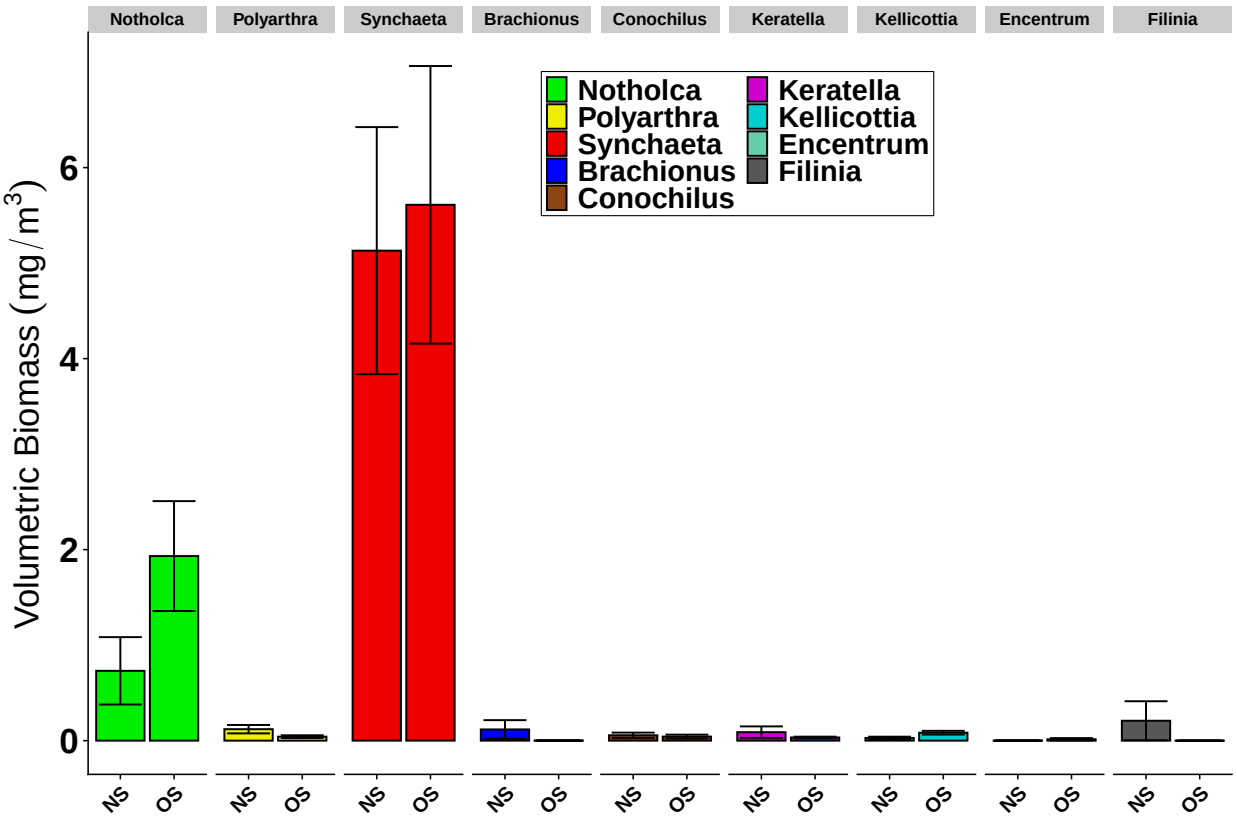


Figure 3: Figure 2A. Spring Rotifer genera nearshore (NS) vs. offshore (OS) average biomass (mg dry wt per m^3). Error bars represent standard errors of the means. Taxa groups are identified in legend by color.

Figure 2A shows the biomass of the 9 most commonly encountered taxa between the nearshore and offshore samples for Lake Erie in April 2024. With several exceptions, most rotifer taxa exhibited higher average biomass values in the offshore samples. Several taxa (*Polyarthra*, *Brachionus*, *Filinia*) showed higher densities in the nearshore vs. offshore waters. *Notholca* was the only taxa that showed a clear preference for the offshore waters. The remaining taxa *Synchaeta*, *Conochilus*, *Keratella*, *Kellicottia*, *Encentrum*, exhibited very similar biomass values between near- and offshore waters. These insights suggest that despite potentially higher food concentrations in nearshore habitats, only several taxa gain an advantage in these conditions.

Nearshore Only: Asplanchna, Mytilina, Ploesoma, Trichocera, Trichotria Offshore Only: Colurella, Gastropus, Monostyla, Pompholyx, Rotifer Bdelloid

18 rotifer genera found in NS spring; 32 speccodes (species) Asplanchna Brachionus Cephalodella Conochilus Encentrum Euchlanis Filinia Kellicottia Keratella Mytilina Notholca Notomatta Ploesoma Polyarthra Proales Synchaeta Trichocerca Trichotria

18 rotifer genera found in the OS spring; 35 speccodes (species) Brachionus Cephalodella Colurella Conochilus Encentrum Euchlanis Filinia Gastropus Kellicottia Keratella Monostyla Notholca Notomatta Polyarthra Pompholyx Proales Rotifer Bdelloid Synchaeta

Rotifer Genera Spring Nearshore-Offshore Density Barplot

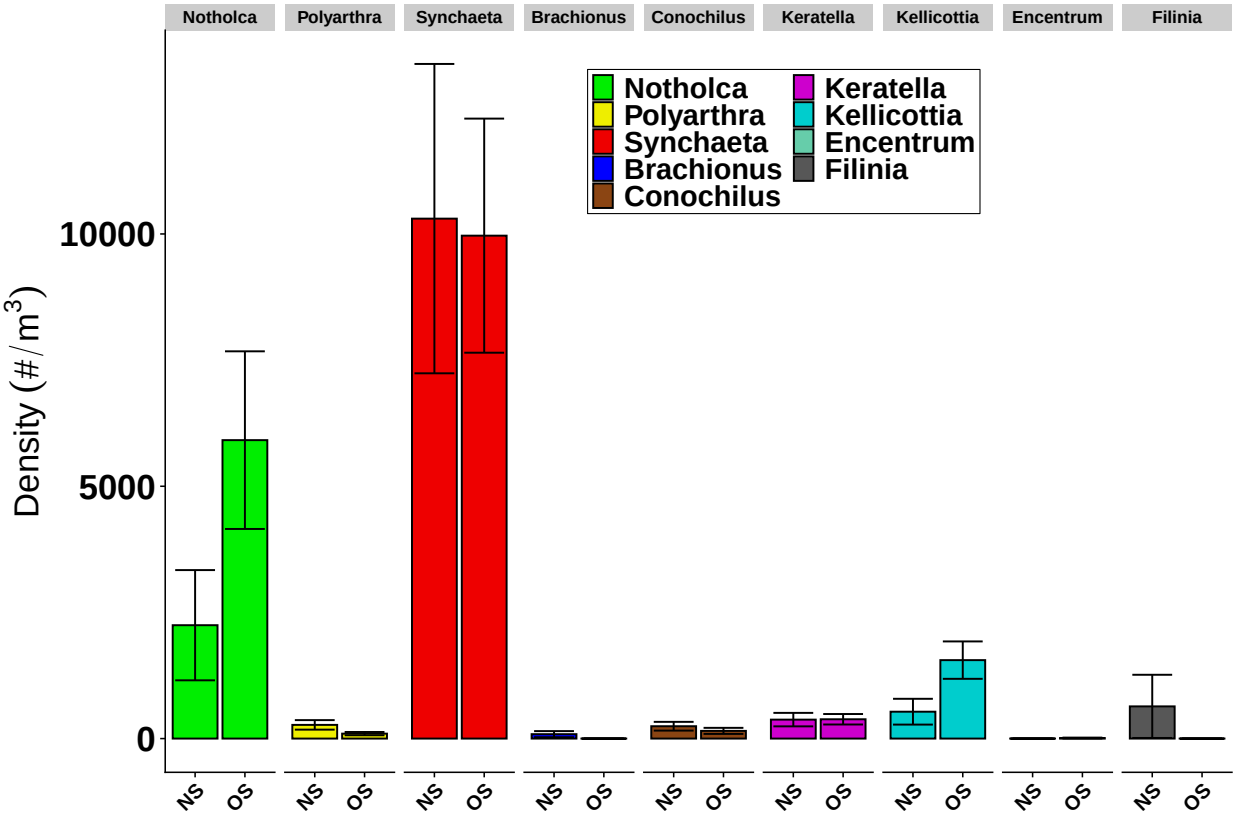


Figure 4: Figure 2B. Spring Rotifer genera nearshore (NS) vs. offshore (OS) average density (# per m^3). Error bars represent standard errors of the means. Taxa groups are identified in legend by color.

Figure 2B shows the density of the 9 most commonly encountered taxa between the nearshore and offshore samples for Lake Erie in April 2024. With several exceptions, most rotifer taxa exhibited higher average density values in the nearshore samples. Several taxa (*Polyarthra*, *Brachionus*, *Filinia*) showed higher densities in the nearshore vs. offshore waters. *Notholca* and *Kellicottia* both showed a clear preference for the offshore waters. The remaining taxa *Synchaeta*, *Conochilus*, *Keratella*, *Encentrum*, exhibited very similar density values between near- and offshore waters. These insights suggest that despite potentially higher food concentrations in nearshore habitats, only several taxa gain an advantage in these conditions.

Rotifer Genera Summer Nearshore-Offshore Biomass Boxplot

Table 1: Table 1. Rotifer Biomass Summer LME Tukey Pairwise Comparisons

contrast	Rot_GENERA	estimate	SE	df	t.ratio	p.value
NS - OS	Ascomorpha	-45.733333	32.2407	596.5741	-1.4184967	0.1565680
NS - OS	Asplanchna	-18.766667	32.2407	596.5741	-0.5820799	0.5607327
NS - OS	Brachionus	-4.033333	32.2407	596.5741	-0.1251007	0.9004860
NS - OS	Collotheca	14.966667	32.2407	596.5741	0.4642165	0.6426620
NS - OS	Conochiloides	-63.550000	32.2407	596.5741	-1.9711108	0.0491730

contrast	Rot_GENERA	estimate	SE	df	t.ratio	p.value
NS - OS	Conochilus	13.900000	32.2407	596.5741	0.4311320	0.6665281
NS - OS	Euchlanis	0.000000	32.2407	596.5741	0.0000000	1.0000000
NS - OS	Filinia	-8.950000	32.2407	596.5741	-0.2775994	0.7814161
NS - OS	Gastropus	-23.500000	32.2407	596.5741	-0.7288923	0.4663536
NS - OS	Kellicottia	0.000000	32.2407	596.5741	0.0000000	1.0000000
NS - OS	Keratella	32.300000	32.2407	596.5741	1.0018392	0.3168275
NS - OS	Lepadella	20.083333	32.2407	596.5741	0.6229186	0.5335761
NS - OS	Mytilina	0.000000	32.2407	596.5741	0.0000000	1.0000000
NS - OS	Ploesoma	39.866667	32.2407	596.5741	1.2365321	0.2167474
NS - OS	Polyarthra	2.733333	32.2407	596.5741	0.0847790	0.9324656
NS - OS	Pompholyx	18.833333	32.2407	596.5741	0.5841477	0.5593418
NS - OS	Proales	0.000000	32.2407	596.5741	0.0000000	1.0000000
NS - OS	Synchaeta	46.200000	32.2407	596.5741	1.4329712	0.1523899
NS - OS	Trichocerca	16.333333	32.2407	596.5741	0.5066060	0.6126185
NS - OS	Trichotria	18.250000	32.2407	596.5741	0.5660546	0.5715695

Rotifer Genera Spring Nearshore-Offshore Biomass Boxplot

Table 2: Table 2. Rotifer Biomass Spring LME Tukey Pairwise Comparisons

contrast	Rot_GENERA	estimate	SE	df	t.ratio	p.value
NS - OS	Brachionus	37.3666667	33.05585	677.1882	1.1304100	0.2587037
NS - OS	Cephalodella	57.2666667	33.05585	677.1882	1.7324213	0.0836539
NS - OS	Collotheca	0.0000000	33.05585	677.1882	0.0000000	1.0000000
NS - OS	Colurella	-17.2250000	33.05585	677.1882	-0.5210877	0.6024760
NS - OS	Conochiloides	0.0000000	33.05585	677.1882	0.0000000	1.0000000
NS - OS	Conochilus	147.0166667	33.05585	677.1882	4.4475228	0.0000102
NS - OS	Encentrum	12.4666667	33.05585	677.1882	0.3771394	0.7061881
NS - OS	Euchlanis	-7.2750000	33.05585	677.1882	-0.2200820	0.8258736
NS - OS	Filinia	103.6583333	33.05585	677.1882	3.1358540	0.0017877
NS - OS	Kellicottia	5.9500000	33.05585	677.1882	0.1799984	0.8572077
NS - OS	Keratella	27.1083333	33.05585	677.1882	0.8200766	0.4124610
NS - OS	Lepadella	0.0000000	33.05585	677.1882	0.0000000	1.0000000
NS - OS	Monostyla	-14.0250000	33.05585	677.1882	-0.4242819	0.6714950
NS - OS	Mytilina	22.9583333	33.05585	677.1882	0.6945315	0.4875871
NS - OS	Notholca	21.3750000	33.05585	677.1882	0.6466328	0.5180887
NS - OS	Notomatta	-4.7250000	33.05585	677.1882	-0.1429399	0.8863802
NS - OS	Polyarthra	24.8000000	33.05585	677.1882	0.7502453	0.4533675
NS - OS	Pompholyx	-13.6750000	33.05585	677.1882	-0.4136937	0.6792294
NS - OS	Proales	35.9083333	33.05585	677.1882	1.0862927	0.2777360
NS - OS	Rotifer Bdelloid	-33.8000000	33.05585	677.1882	-1.0225118	0.3069039
NS - OS	Synchaeta	0.0333333	33.05585	677.1882	0.0010084	0.9991957
NS - OS	Trichocerca	24.2083333	33.05585	677.1882	0.7323463	0.4642106
NS - OS	Trichotria	22.8750000	33.05585	677.1882	0.6920106	0.4891679

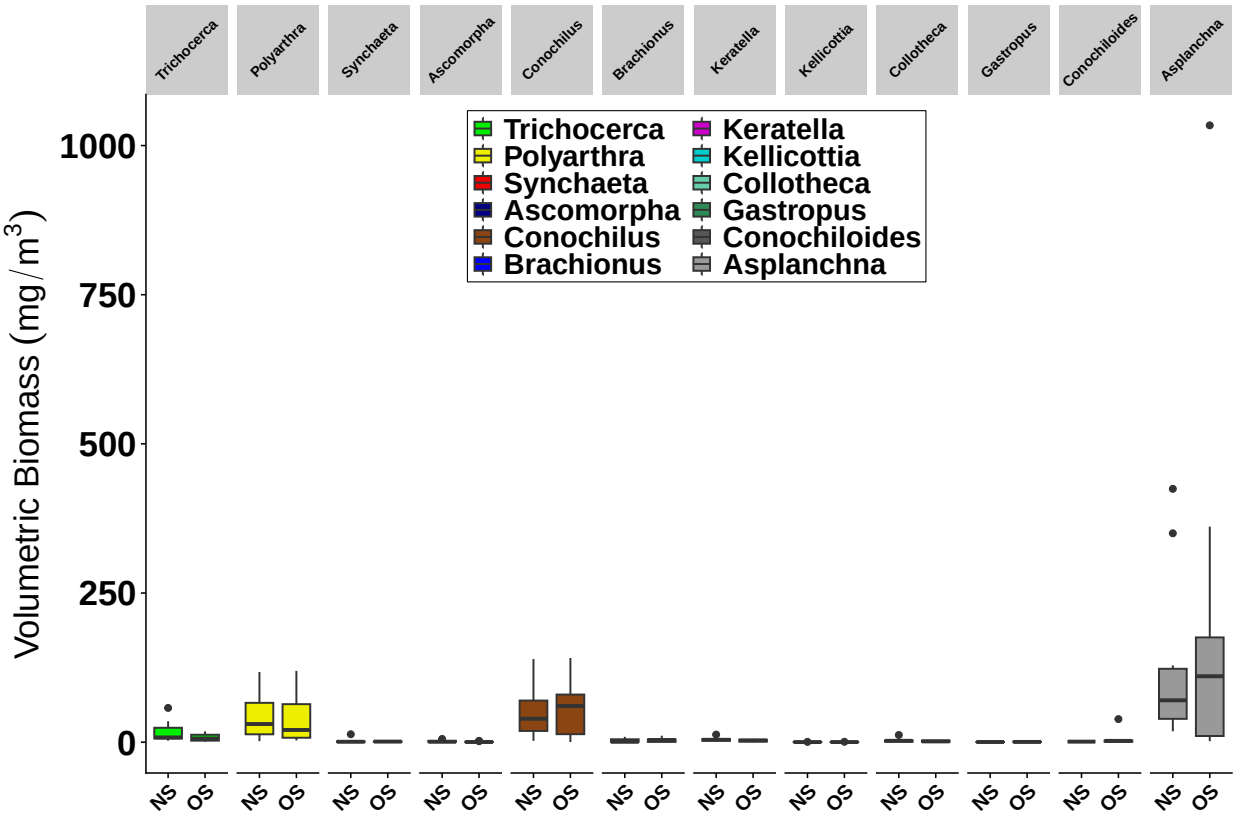


Figure 5: Figure 3A. Summer Rotifer genera nearshore (NS) vs. offshore (OS) average biomass (mg dry wt per m^3) boxplots. Taxa groups are identified in legend by color.

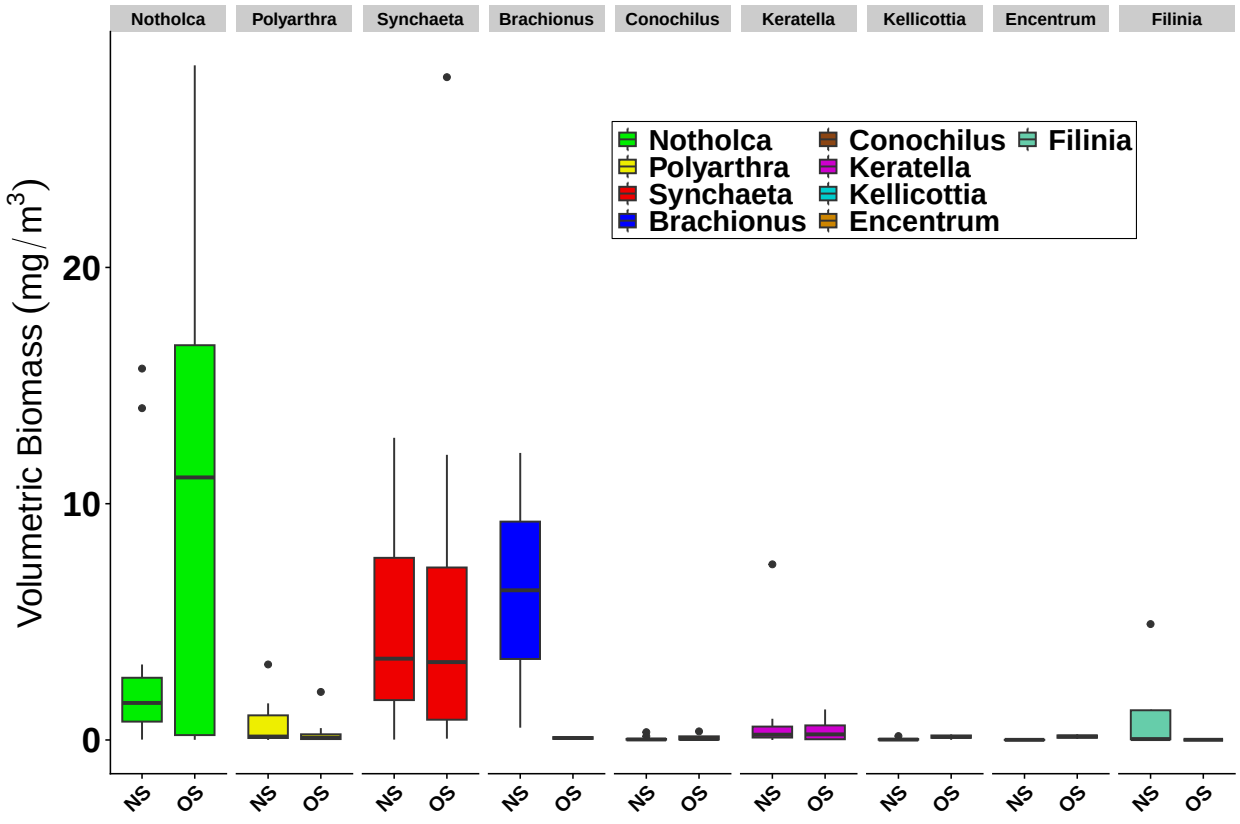


Figure 6: Figure 3B. Spring Rotifer genera nearshore (NS) vs. offshore (OS) average biomass (mg dry wt per m^3) boxplots. Taxa groups are identified in legend by color.

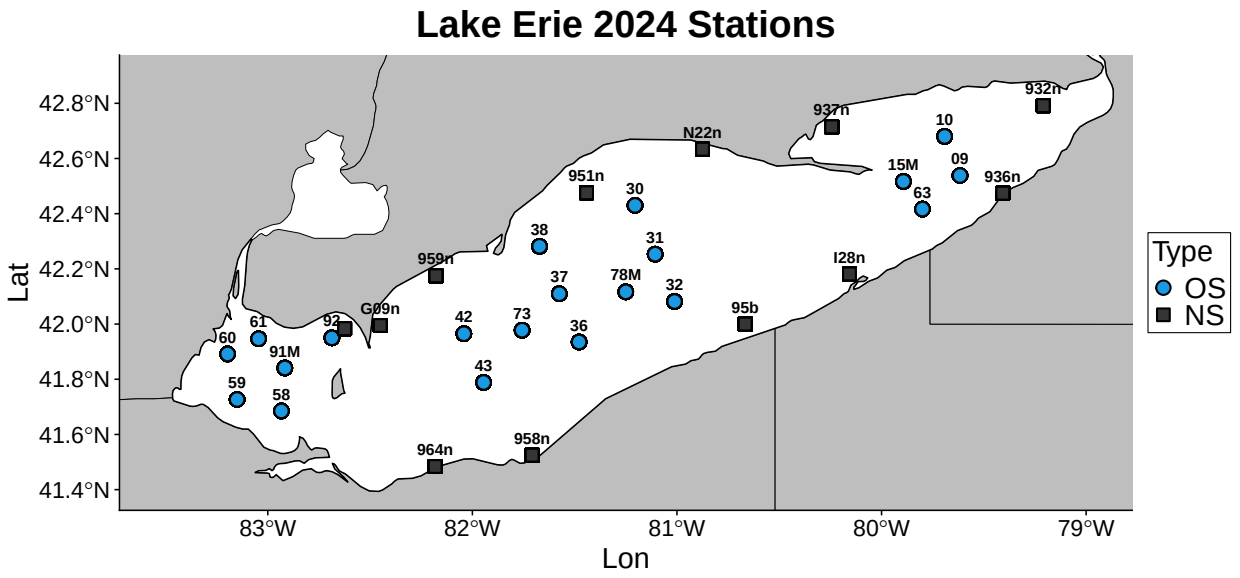


Figure 7: Figure 4. Rotifer study area for 2024 samples collected at the 20 offshore sites (OS - blue filled circles), and the 12 CSMI nearshore sites (NS - black filled boxes).

Lake Erie 2024 Study Area Station Map

Rotifer Summer Genera Station Total Biomass Barplot

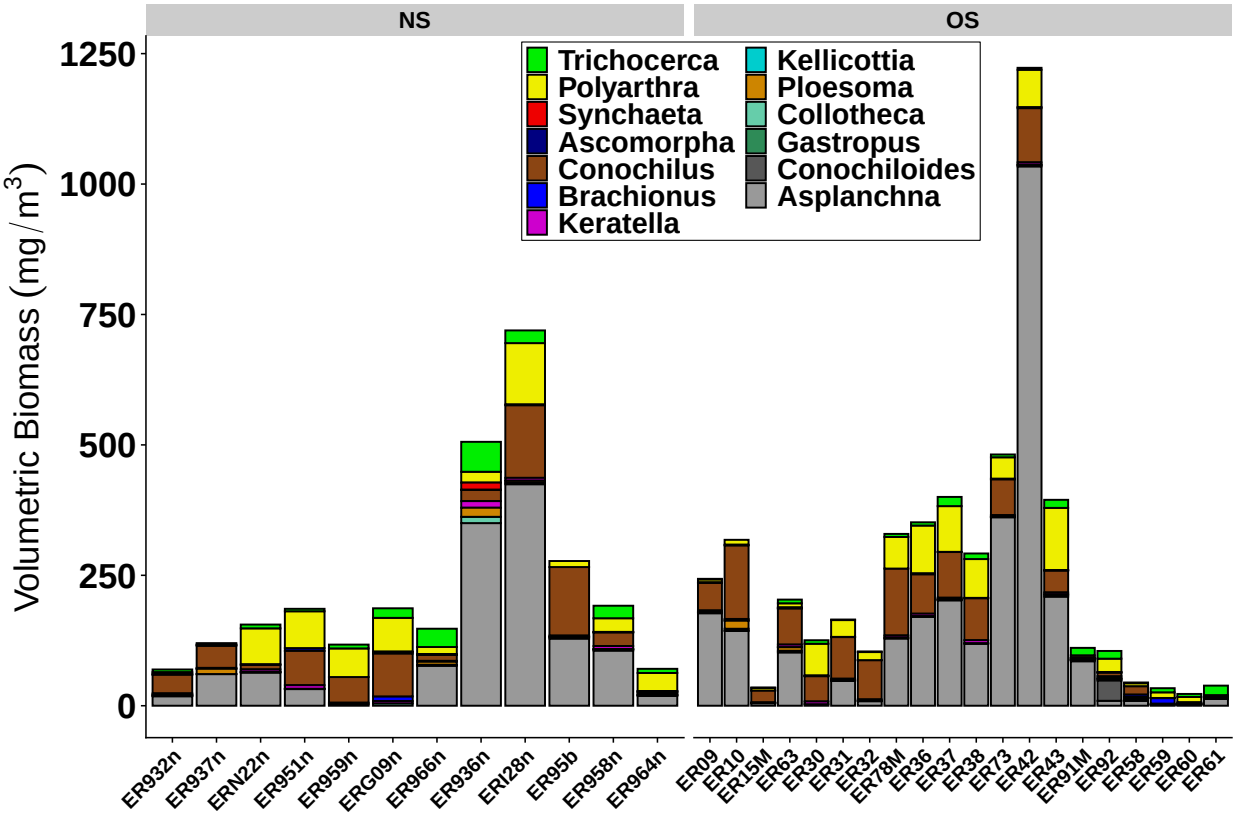


Figure 8: Figure 5A. Summer Rotifer genera station total biomass (mg dry wt per m^3) for the the 12 CSMI nearshore sites ordered from east to west (7 northshore, 5 south), and the 20 offshore sites ordered from east to west (4 east basin, 10 central, 6 west).

Figure 5A represents the individual station total rotifer biomass for the 13 most common taxa collected during the Lake Erie August 2024 survey. The stations are separated by the nearshore (NS) and offshore (OS) samples ordered from east to west. The first 7 listed nearshore samples were collected from the northshore of Lake Erie and show a lower overall biomass compared with the 5 subsequent stations collected from the southshore of the lake. That biomass disparity is largely driven by higher *Asplanchna* values in several southshore samples. Station ER936n shows a particularly diverse community compared with other stations, and is close in proximity to Van Buren Bay close to Dunkirk, NY. The offshore samples are separated by their basins (4 east, 10 central, 6 west), which are distinct due to the large change in bathymetric depths between basins, resulting in very different ecological implications for the zooplankton communities. The eastern basin is the deepest and most similar to the Lake Ontario rotifer community, and with the exception of ER15M which is missing the large *Asplanchna* biomass, those 4 samples are similar. The first 3 central samples are similar with relatively low overall biomass with the addition of more *Polyarthra*. With the exception of ER42 which shows a massive *Asplancha* biomass, the remaining central stations exhibit a similar community, with more *Polyarthra* and *Trichocera* biomass moving westward. The western basin is historically the most diverse community in the Great Lakes, with very shallow and eutrophic waters that support a unique assemblage of rotifers. A higher proportion of *Trichocera*, *Polyarthra*, and *Brachionus* is indication of greater nutrient availability, and therefore a more diverse set of food resources compared with other localities. It should be noted that the despite these resources, the western basin often exhibits the lowest biomass and density of

rotifers, which is impacted by the shallow waters (~6 meters), discharge from the Detroit River and Maumee River, and higher potential of predation by larger zooplankton and larval fish.

Rotifer Summer Genera Station Total Density Barplot

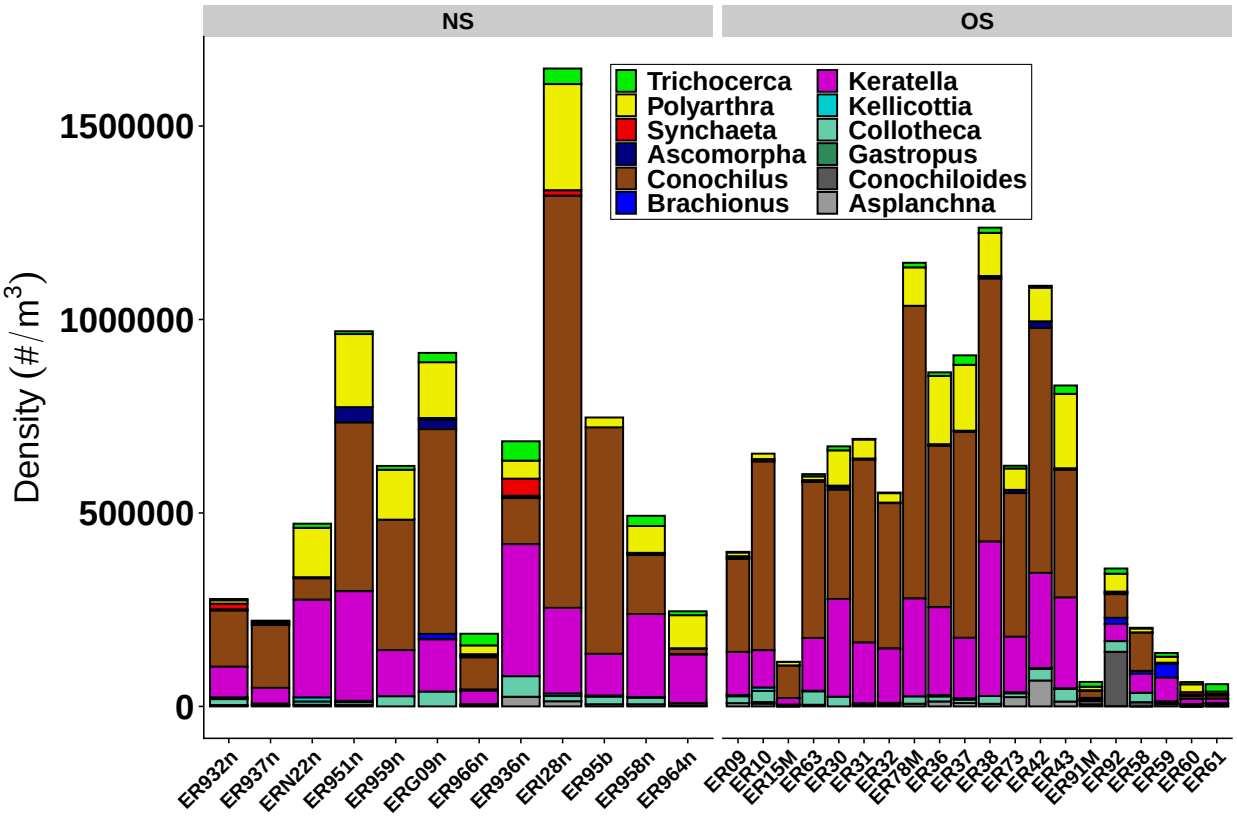


Figure 9: Figure 5B Summer Rotifer genera station total density (# per m^3) for the 12 CSMI nearshore sites ordered from east to west (7 northshore, 5 south), and the 20 offshore sites ordered from east to west (4 east basin, 10 central, 6 west).

Figure 5B represents the individual station total rotifer density for the 13 most common taxa collected during the Lake Erie August 2024 survey. The stations are separated by the nearshore (NS) and offshore (OS) samples ordered from east to west. The first 7 listed nearshore samples were collected from the northshore of Lake Erie and show a similar overall density compared with the 5 subsequent stations collected from the southshore of the lake. That density similarity is driven by several taxa including the colonial *Conochilus*, *Keratella*, and *Polyarthra*.

Working from here

Station ER936n shows a particularly diverse community compared with other stations, and is close in proximity to Van Buren Bay close to Dunkirk, NY. The offshore samples are separated by their basins (4 east, 10 central, 6 west), which are distinct due to the large change in bathymetric depths between basins, resulting in very different ecological implications for the zooplankton communities. The eastern basin is the deepest and most similar to the Lake Ontario rotifer community, and with the exception of ER15M which is missing the large *Asplanchna* biomass, those 4 samples are similar. The first 3 central samples are similar with relatively low overall biomass with the addition of more *Polyarthra*. With the exception of ER42 which shows a massive *Asplancha* biomass, the remaining central stations exhibit a similar community, with more *Polyarthra* and *Trichocera* biomass moving westward. The western basin is historically the most diverse

community in the Great Lakes, with very shallow and eutrophic waters that support a unique assemblage of rotifers. A higher proportion of *Trichocera*, *Polyarthra*, and *Brachionus* is indication of greater nutrient availability, and therefore a more diverse set of food resources compared with other localities. It should be noted that the despite these resources, the western basin often exhibits the lowest biomass and density of rotifers, which is impacted by the shallow waters (~6 meters), discharge from the Detroit River and Maumee River, and higher potential of predation by larger zooplankton and larval fish.

Rotifer Spring Genera Station Total Biomass Barplot

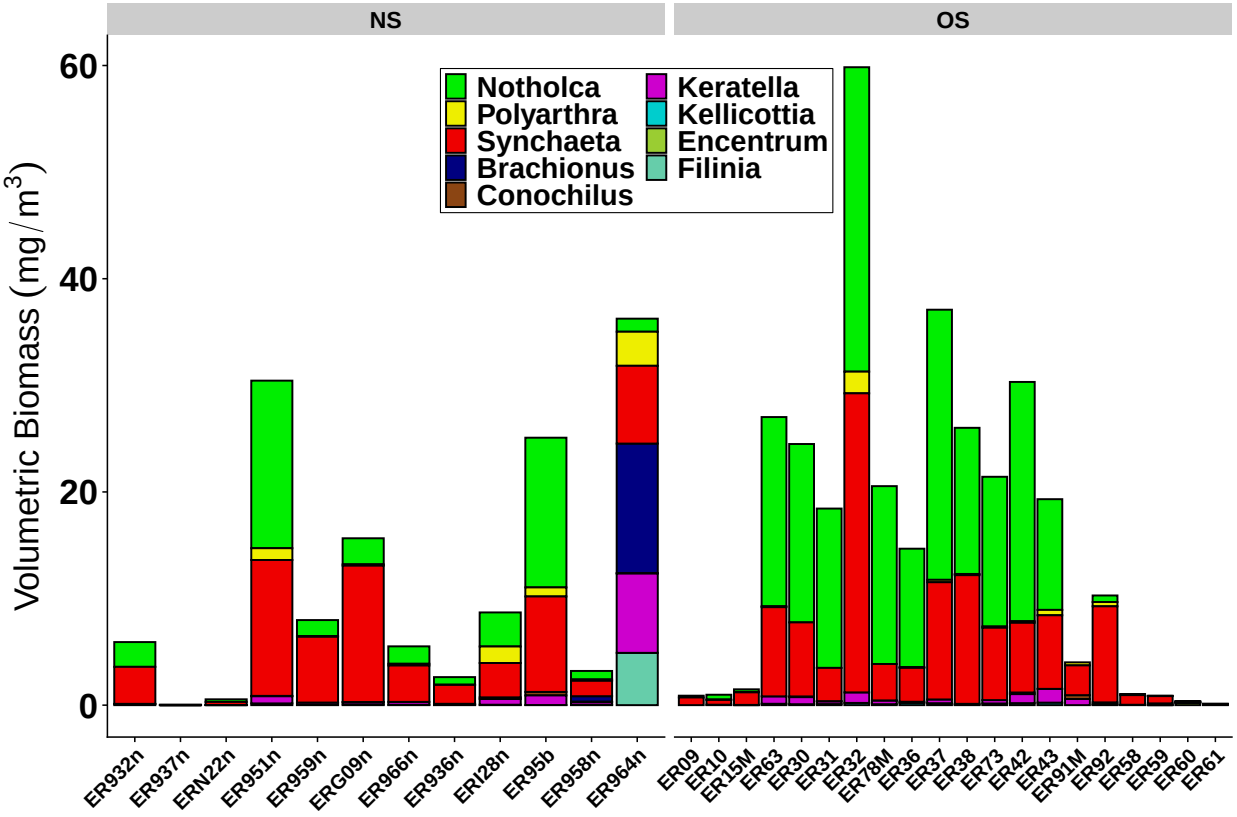


Figure 10: Figure 6A. Spring Rotifer genera station total biomass (mg dry wt per m^3) for the 12 CSMI nearshore sites ordered from east to west (7 northshore, 5 south), and the 20 offshore sites ordered from east to west (4 east basin, 10 central, 6 west).

Rotifer Spring Genera Station Total Density Barplot

Zoop Taxa Summer Nearshore-Offshore Biomass Barplot

Zoop Taxa Spring Nearshore-Offshore Biomass Barplot

Zoop Summer Taxa Station Total Biomass Barplot

Zoop Spring Taxa Station Total Biomass Barplot

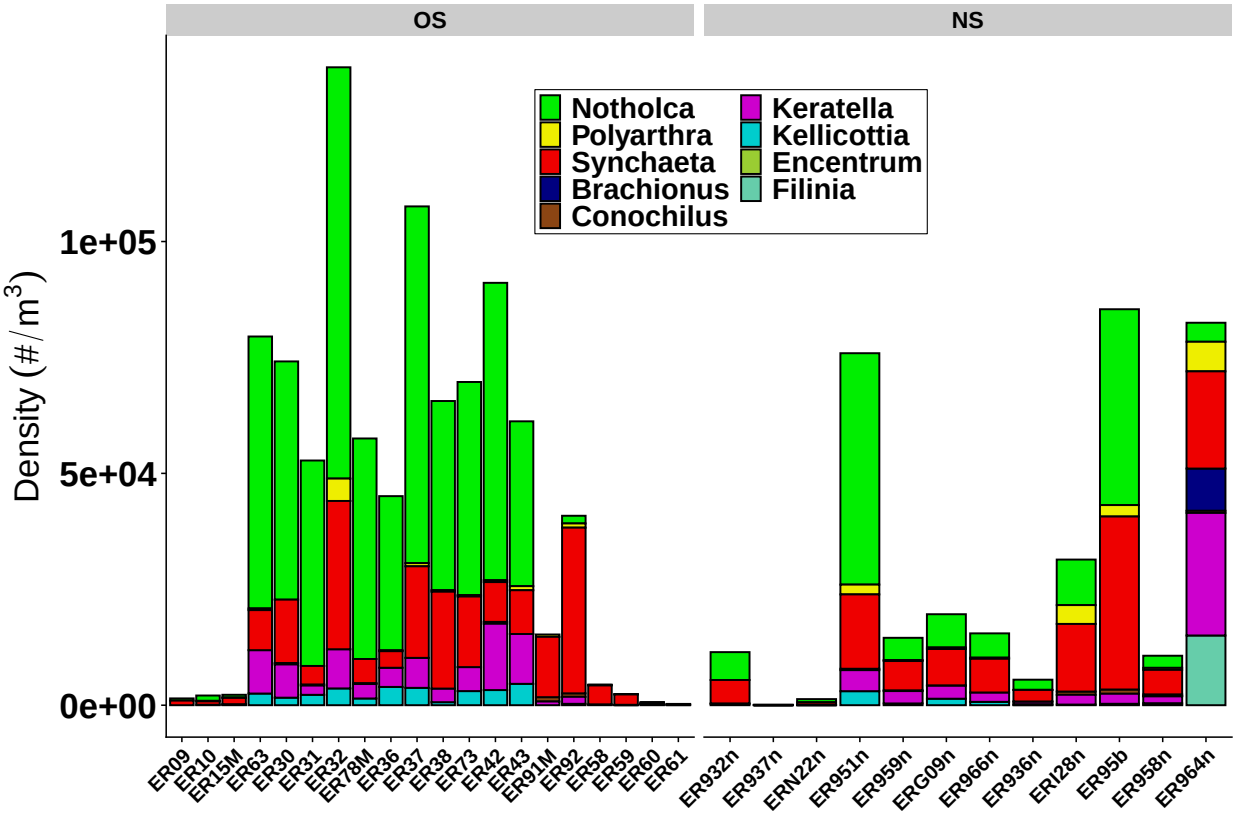


Figure 11: Figure 6B. Spring Rotifer genera station total density (# per m^3) for the 12 CSMI nearshore sites ordered from east to west (7 northshore, 5 south), and the 20 offshore sites ordered from east to west (4 east basin, 10 central, 6 west).

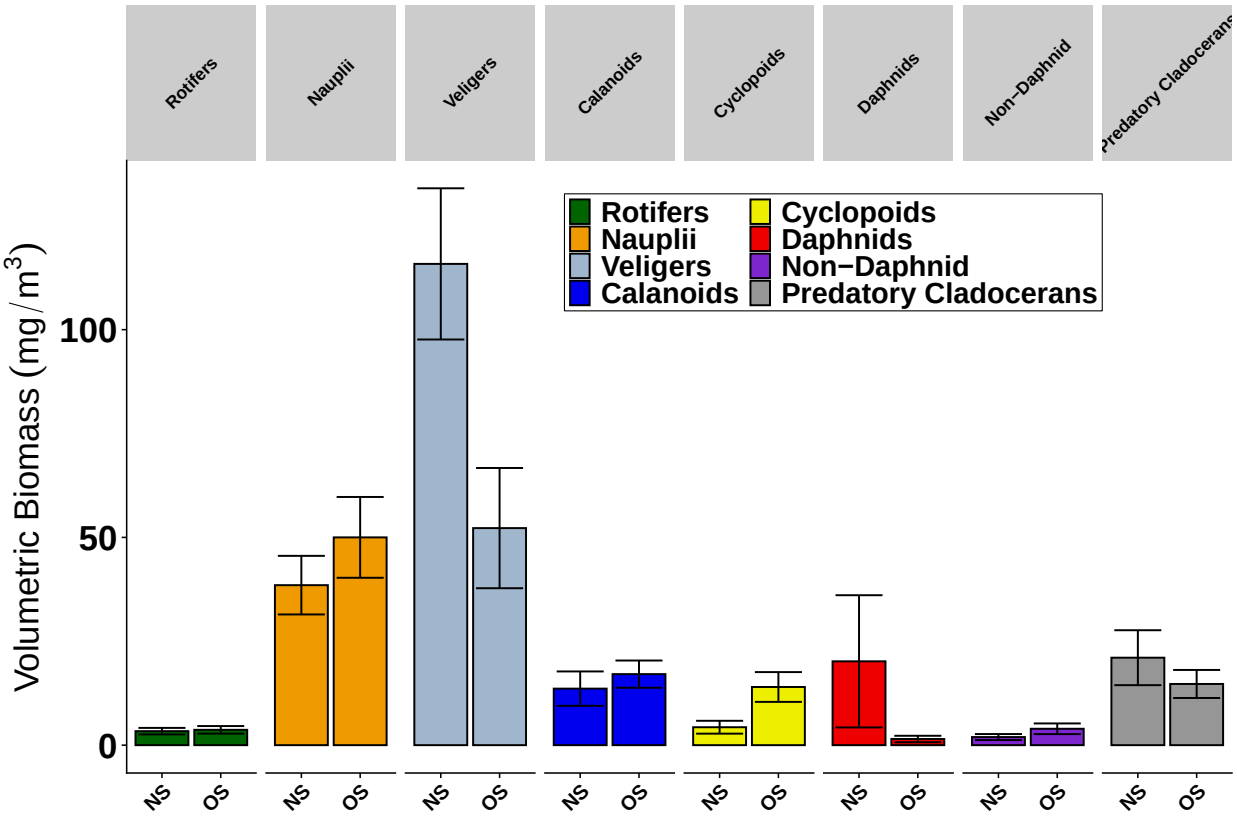


Figure 12: Figure 7A. Summer Zoop taxa nearshore (NS) vs. offshore (OS) average biomass (mg dry wt per m^3). Error bars represent standard errors of the means. Taxa groups are identified in legend by color.

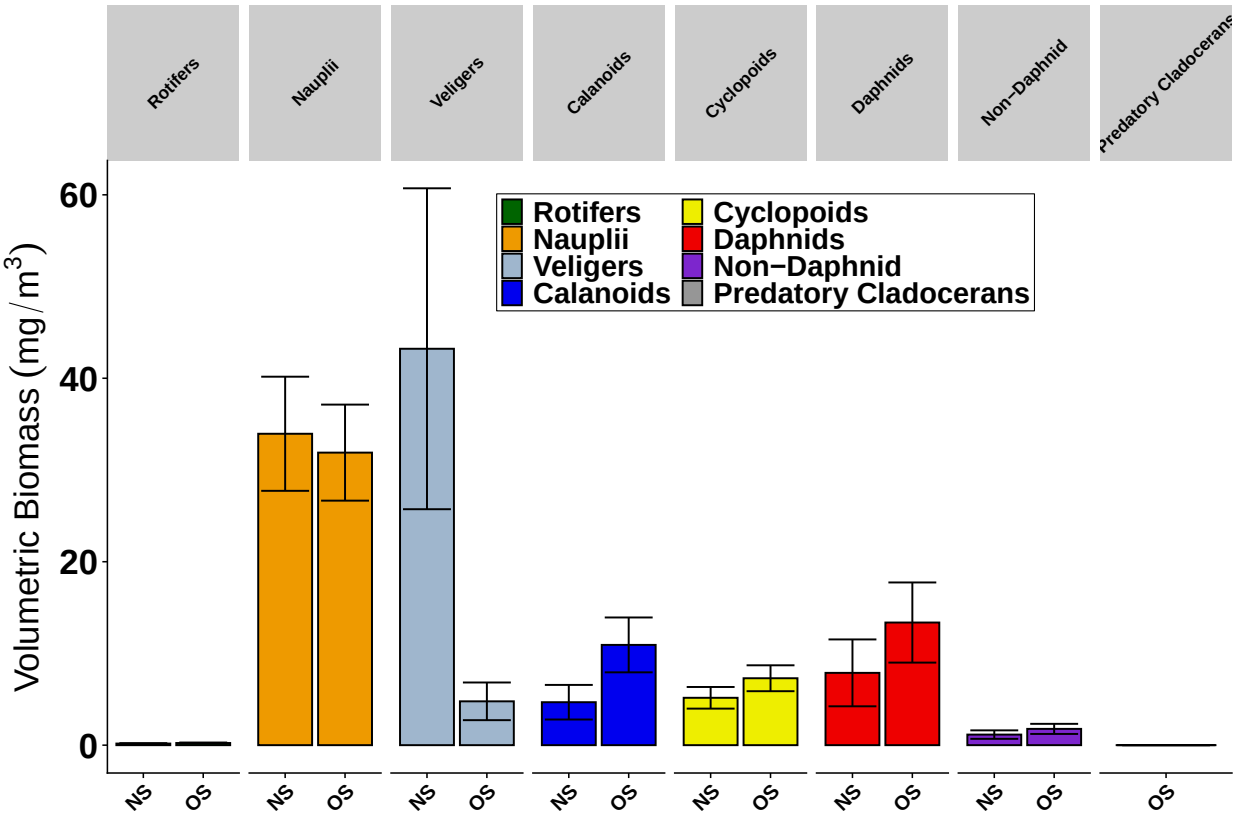


Figure 13: Figure 7B. Spring Zoop taxa nearshore (NS) vs. offshore (OS) average biomass (mg dry wt per m^3). Error bars represent standard errors of the means. Taxa groups are identified in legend by color.

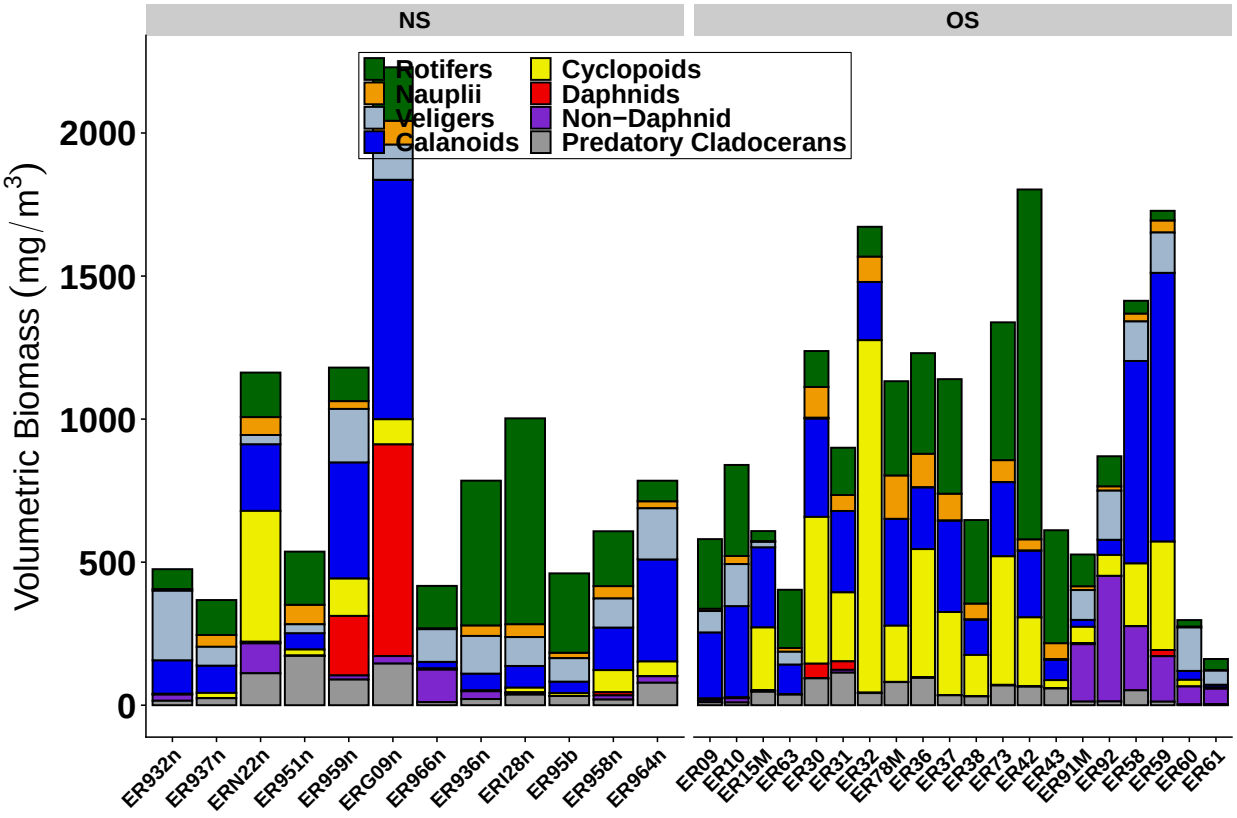


Figure 14: Figure 9A. Summer Zoop taxa station total biomass (mg dry wt per m^3) for the 12 CSMI nearshore sites ordered from east to west (7 northshore, 5 south), and the 20 offshore sites ordered from east to west (4 east basin, 10 central, 6 west).

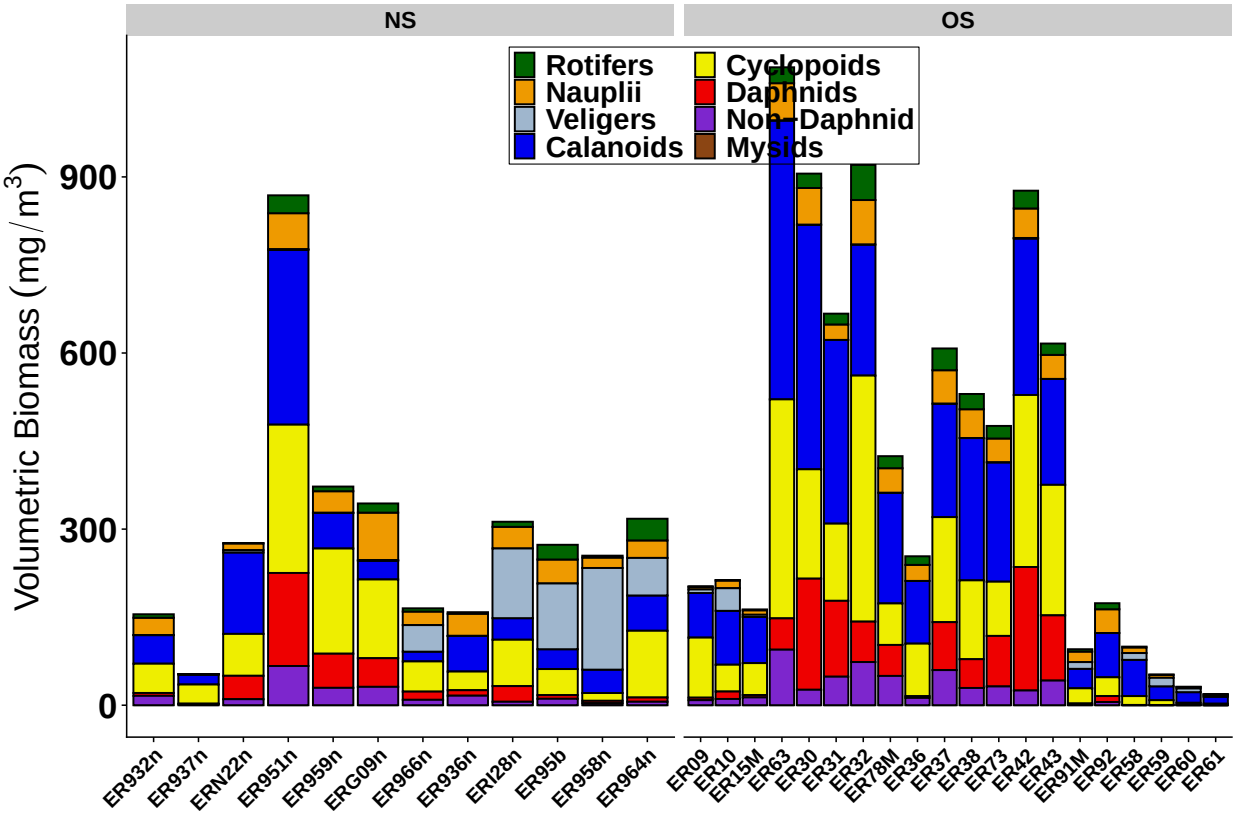


Figure 15: Figure 9B. Spring Zoop taxa station total biomass (mg dry wt per m^3) for the 12 CSMI nearshore sites ordered from east to west (7 northshore, 5 south), and the 20 offshore sites ordered from east to west (4 east basin, 10 central, 6 west).